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Assignment #2

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Data Science (A)

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Course :~

Datal Structure

Submitted to:~

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Data Structure & Algorithm

"Algorithm"

★ Algorithm of Sum:

```
Algorithm Sum (A,n)
{
    S=0;        — 1
for (i=0; i<n; i++) {   — n+1
    S = S + A[i];       — n
}
    return S;           — 1
}
```

Time Complexity:-

$f(n) = 2n+3$

$O(n)$

Space Complexity:-

A — n

n = 1

S = 1

i = 1

$S(n) = n+3$

$O(n)$

Algorithm : Sum of matrices

```
Algorithm (A, B, n)
{
```

```
for (i=0; i<n; i++)  — n+1
{
  for (j=0; j<n; j++)  — n*(n+1)
  {
    C[i,j] = A[i,j] + B[i,j]; — n*n
  }
}
```

Time Complexity:

$f(n) = 2n^2 + 2n + 1$

$O(n^2)$

Space Complexity:

A - n^2

B - n^2

C - n^2

n - 1, i - 1, j - 1

$S(n) = 3n^2 + 3$

$O(n^2)$

Algorithm of Multiplication

```
Algorithm Multiply (A, B, n)
{
  for (i=0; i<n; i++)  — n+1
  {
    for (j=0; j<n; j++)  — n(n+1)
    {
      C[i,j] = 0  — n*n
    for (k=0; k<n; k++)  — (n+1) * n*n
    {
      C[i,j] = C[i,j] + A[i,k] * B[k,j]; — n*n
```

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Time Complexity:

$f(n) = 2n^3 + 3n^2 + 2n + 1$

$O(n^3)$

Space Complexity:

A - n^2

B - n^2

C - n^2

n - 1 , i - 1 , j - 1 , k - 1

$S(n) = 3n^2 + 4$

$O(n^2)$

Time Complexity

(i)

```
for (i=1; i<n; i+=20) — n+1
{
    Stmt — n/2
}
```

Time Complexity:

$f(n) = n/20$

$O(n)$

(ii)

```
for (i=0; i<n; i++) — n+1
{
    for (j=0; j<n; j++) — n*(n+1)
```

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```
{
    stmt; —— n x n
}
```

$O(n^2)$

(iii)

```
for (i=0; i<n; i++)
{
    for (j=0; j<i; j++)
    {
        stmt;
    }
}
```

$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$

$f(n) = n+1/2$

$O(n^2)$

(iv)

```
P=0
for (i=1; P<n; i++) {
    P=P+i;
}
```

Assume

$\therefore P = k(k+1)/2$

$k(k+1)/2 > n$

$k^2 > n$

$k > \sqrt{n}$

$O(\sqrt{n})$

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(v)

```
for (i=1; i<n; i=i*2)
{
    Stmt;
}
```

Assume that $i >= n$

$i = 2^k$

$2^k >= n$

if $2^k = n$

then $k = \log_2 n$

(vi)

```
for (i=1; i<=n; i++)
{
    Stmt;
}
```

$i = 1+1+1+\cdots+1 = n$

$k = n$

$O(n)$

(vii)

```
for (i=n; i>=1; i=n/2)
{
    Stmt;
}
```

Assume: k is < 1

$\frac{n}{2^k} < 1$

$\frac{n}{2^k} = 1$

$n = 2^k$

$k = \log_2 n$

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( )

```
for (i=0; i*i<n; i++)
{
    stmt;
}
```

i*i<n

i*i>n

i>n

i = √n => O(√n)

( )

```
for (i=0; i<n; i++)
{
    stmt
}
for (j=0; j<n; j++)
{
    stmt;
}
```

=> O(n)

( )

```
p=0
for (i=1; i<n; i = i*2)
{
    p++;
}
for (j=1; j<p; j = j*2)
{
    stmt;
}
```

p = log n

=> log log n => O(log n log n)

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( )

```
for (i=0; i<n; i++)
{
    for (j=0; j<n; j*2)
    {
        stmt;  —— n x logn
    }
}
```

⇒ 2n logn + n

O(n log 2)

(Overview)

1. for (i=0; i<n; i++) —— O(n)
2. for (i=0; i<n; i+2) —— O(log₂ n)
3. for (i=n; i>1; i--) —— O(log₂n)
4. for (i=1; i<n; i*2)
   for (i=1; i<n; i=i*3)
   for (i=n; i>1; i=i/2)
)

Date:

(Overview)

for (i=0; i<n; i++) — O(n)

for (i=0; i<n; i+2) — $O(\log_2 n) \simeq O(n)$

for (i=n; i>1; i--) — $O(\log_3 n)$ O(n)

for (i=1; i<n; i*2) — $O(\log_2 n)$

for (i=1; i<n; i=i*3) — $O(\log_3 n)$

for (i=n; i>1; i=i/2) — $O(\log_2 n)$

Analysis for if and while

①

```
i = 0          —— 1
while (i<n)    —— n+1
{
    stmt;      —— n
    i++;       —— n
}
```

$f(n) = 3n+2$

$\Rightarrow O(n)$

②

```
a = 1
while (a<b)
{
    stmt;
    a = a*2;
}
```

③

```
i = n
while (i>1)
{
    stmt;
    i = i/2;
}
```

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Date:

Terminate

$a \geq b$

$a = 2^k$

$2^k \geq b$

$2^k = b$

$k = \log_2 b$

$\Rightarrow O(\log n)$

$\Rightarrow O(\log_2 n)$

④

```
i = 1;
k = 1;
while (k < n)
{ stmt;
  k = k + i;
  i++;
}
```

i	k
1	1
2	1+1=2
3	2+2 =
4	2+2+3
5	2+2+3+4
.	.
.	.
m	2+2+2+4+...+m

$\therefore \frac{m(m+1)}{2}$

Assume

$k \geq n$

$\frac{m(m+1)}{2} > m$

$m^2 \geq m$

$m = \sqrt{n}$

$\Rightarrow O(\sqrt{n})$

⑤

```
while (m != n)
{
  if (m > n)
    m = m - n;
  else
    n = n - m;
}
```

m	n
12	2
10	2
8	2
6	2
4	2
2	2

$\Rightarrow O(n)$

$\Rightarrow \min O(1)$

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4. Set Temp = A[U]
5. Set LOC = U
6. Set I = U+1
7. Repeat Step 8 to 9 while (I <= N)
8. If Temp > A[I] then
   Temp = A[I]
   LOC = I
   End If
9. I = I+1
10. T = A[LOC]
    A[LOC] = A[U]
11. U = U+1
12. Exit.

Dry Run:-

Initial step:

- Array A = [64, 25, 12]
- Size N = 3
- I. Start

Pass 1 (U = 1)

- Set U = 1

Check condition (U < N) ⇒ (1 < 3) True → Proceed

Set Temp = A[1] = 64

Set LOC = U = 1

Set I = U+1 = 2

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Inner Loop: (find Max in unsorted portion):

Check (I<N) ⇒ (2 < 3) True

Compare Temp ≠ A[2] ⇒ 64 > 25 (True)

Temp = A[2] = 25

LOC = 2

I = I+1 = 3

Check (I≤N) ⇒ (3≤3) (True):

Compare Temp > A[3] ⇒ 25 > 12 (True)

Temp = A[3] = 12

LOC = 3

I = I+1 = 4

Check (I ≤ N) ⇒ (4≤3) (False):

Exit inner loop.

Swap Minimum to Position U:-

Swap A[1] & A[LOC]

T = A[LOC] = A[3] = 12

A[LOC] = A[3] = A[1] = 64

A[1] = T = 12

Array status after Pass 2 = A = [12, 25, 64]

Increment U = U+1 = 3

Loop Termination:

Check condition (U<N) ⇒ (3<3)

1 false → Exit main loop

Exit

Final result

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Algorithm test:

```
Algorithm test(n)
{
    if(n<5)
    {
        printf("%d", n);
    }
else
    {
    for(i=0; i<n; i++)
        {
        printf("%d", i);
        }
    }
}
```

Best — O(1)
Worst — O(n)

Algorithm — Selection Sort

Selection Sort()

Algorithm to sort an array A consisting of N elements in ascending.

1. Start
2. Set V=1
3. Repeat Step 4 to 11 while(V<N)

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Algorithm - Bubble Sorting

Bubble (DATA, N)

Here DATA, is an array with N elements.

This algorithm sorts the element in Data

1. Repeat Step 2 & 3 for K = 1 to N-1
2. Set PTR := 1 [Initialize pass pointer PTR]
3. Repeat while PTR <= N-K, [execute pass]
   1. if DATA [PTR] > DATA [PTR+1], then
      Interchange DATA [PTR] and DATA [PTR+1]
      [end of if structure]
   2. Set PTR = PTR+1
      [end of inner loop]
   3. [end of step 2 outer loop]
4. exit.

Dry Run:

Initial Step:

Array : DATA = [34, 11, 20]

Size N=3

Pass 1 (K=1)

Outer loop starts with K=1

PTR = 1

Check Inner loop condition (PTR < N-K)

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=> (1 ≤ 3 - 1) => (1 < 2) (True)

Iteration 1 (PTR = 1)

Compare DATA[1] > DATA[2] => 34 > 15 (True)

Interchange DATA[1] and DATA[2]

Array becomes DATA = [15, 34, 20]

PTR = 1 + 1 = 2

Iteration 2 (PTR = 2)

Check (PTR ≤ 2) => (2 ≤ 2) (True)

Compare DATA[2] > DATA[3] => 34 > 20 (True)

Interchanging DATA[2] and DATA[3]

Array becomes DATA = [15, 30, 34]

PTR = 2 + 1 = 3

Inner Loop Exit:

Check (PTR < 2) => (3 ≤ 2) (false → Exit inner loop)

Array status after Pass 1: [15, 20, 34]

(largest element 34 is not at its correct final position)

Pass (K = 2)

Outer loop advances to K = 2

PTR = 1

Check inner loop condition (PTR ≤ N - K)

=> (1 ≤ 3 - 2) => (1 ≤ 1) (True)

Iteration 1 (PTR = 1)

Compare DATA[1] > DATA[2] => 15 > 20 (False)

No swap needed

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PTR = 1+1 = 2

Inner loop exit

Check (PTR ≤ 1) ⇒ (2 < 1) (false → Exit Inner loop)

Array status after pass 2 = [15, 20, 24]

Loop Termination.

Outer loop check (K ≤ N − 1) ⇒ (3 ≤ 2) (False → Exit outer loop)

Exit

Final Result:-

Sorted Array DATA [15, 20, 34]

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